

Chapter 4

Ethical and Social Issues in Information Systems

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LEARNING OBJECTIVES

- **Identify the ethical, social, and political issues that are raised by information systems.**
- **Identify the principles for conduct that can be used to guide ethical decisions.**
- **Evaluate the impact of contemporary information systems and the Internet on the protection of individual privacy and intellectual property.**
- **Assess how information systems have affected everyday life.**

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Is Your Student Loan Data on Loan?

- **Problem: Insufficient privacy protections for sensitive data related to student loans.**
- **Solutions: Improve system security and protect student information to restore confidence in the system.**
- **Revoke over 52,000 user IDs suspected of misusing access to students' private information.**
- **Demonstrates IT's role in providing quick and convenient access to data.**
- **Illustrates how the very same technology has the potential to threaten privacy and cause more harm than good.**

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Understanding Ethical and Social Issues Related to Systems

- **Past 10 years: One of the most ethically challenged periods in U.S. history**
  - Lapses in management ethical and business judgment in a broad spectrum of industries
  - Enron, WorldCom, Parmalat, etc.
  - Sub-prime loans and the failure of risk analysis: CitiBank and Societe Generale
  - Information systems instrumental in many recent frauds
  - Stiffer sentencing guidelines, obstruction charges against firms, mean individual managers must take greater responsibility regarding ethical and legal conduct

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TABLE 4.1 RECENT EXAMPLES OF FAILED ETHICAL JUDGMENT BY SENIOR MANAGERS

Barclays Bank PLC (2012)	One of the world's largest banks admitted to manipulating its submissions for the LIBOR benchmark interest rates in order to benefit its trading positions and the media's perception of the bank's financial health. Fined $160 million.
GlaxoSmithKline LLC (2012)	The global health care giant admitted to unlawful and criminal promotion of certain prescription drugs, its failure to report certain safety data, and its civil liability for alleged false price reporting practices. Fined $3 billion, the largest health care fraud settlement in U.S. history and the largest payment ever by a drug company.
Walmart Inc. (2012)	Walmart executives in Mexico accused of paying millions in bribes to Mexican officials in order to receive building permits. Under investigation by the Department of Justice.
Minerals Management Service (U.S. Department of the Interior) (2010)	Government managers accused of accepting gifts and other favors from oil companies, letting oil company rig employees write up inspection reports, and failing to enforce existing regulations on offshore Gulf drilling rigs. Employees systematically falsified information record systems.
Pfizer, Eli Lilly, and AstraZeneca (2009)	Major pharmaceutical firms paid billions of dollars to settle U.S. federal charges that executives fixed clinical trials for antipsychotic and pain killer drugs, marketed them inappropriately to children, and claimed unsubstantiated benefits while covering up negative outcomes. Firms falsified information in reports and systems.
Galleon Group (2011)	Founder of the Galleon Group sentenced to 11 years in prison for trading on insider information. Found guilty of paying $250 million to Wall Street banks, and in return received market information that other investors did not get.
Siemens (2009)	The world's largest engineering firm paid over $4 billion to German and U.S. authorities for a decades-long, worldwide bribery scheme approved by corporate executives to influence potential customers and governments. Payments concealed from normal reporting accounting systems.
IBM (2011)	IBM settled SEC charges that it paid off South Korean and Chinese government officials with bags of cash over a 10-year period.
McKinsey & Company (2011)	CEO Rajat Gupta heard on tapes leaking insider information. The former CEO of prestigious management consulting firm McKinsey & Company was found guilty in 2012 and sentenced to two years in prison.
Tyson Foods (2011)	World's largest producer of poultry, beef, and pork agreed to pay $5 million in fines for bribing Mexican officials to ignore health violations.

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- **Ethics**
  - Principles of right and wrong that individuals, acting as free moral agents, use to make choices to guide their behavior
- **Information systems and ethics**
  - Information systems raise new ethical questions because they create opportunities for:
    - Intense social change, threatening existing distributions of power, money, rights, and obligations
    - New kinds of crime

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- **A model for thinking about ethical, social, and political issues**
  - Society as a calm pond
  - IT as a rock dropped in pond, creating ripples of new situations not covered by old rules
  - Social and political institutions cannot respond overnight to these ripples — it may take years to develop etiquette, expectations, laws
  - Requires understanding of ethics to make choices in legally gray areas

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The Relationship Between Ethical, Social, and Political Issues in an Information Society

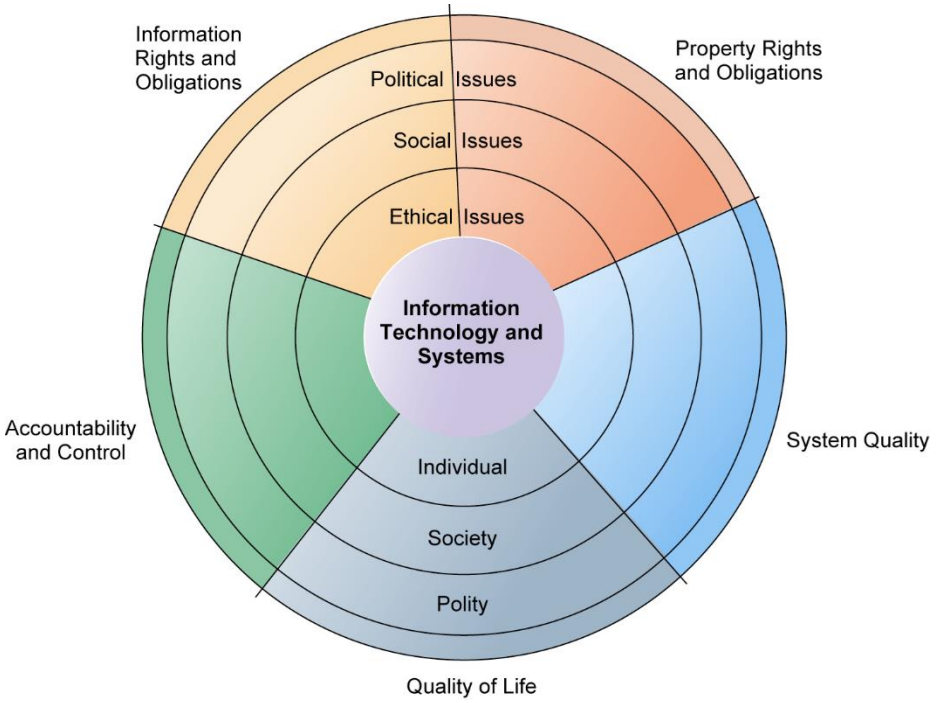


Figure 4-1

The introduction of new information technology has a ripple effect, raising new ethical, social, and political issues that must be dealt with on the individual, social, and political levels. These issues have five moral dimensions: information rights and obligations, property rights and obligations, system quality, quality of life, and accountability and control.

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- **Five moral dimensions of information age**
  - **Major issues raised by information systems include:**
    - Information rights and obligations
    - Property rights and obligations
    - Accountability and control
    - System quality
    - Quality of life

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- **Four key technology trends that raise ethical issues**
  - **Computing power doubles every 18 months**
    - Increased reliance on, and vulnerability to, computer systems
  - **Data storage costs rapidly declining**
    - Multiplying databases on individuals
  - **Data analysis advances**
    - Greater ability to find detailed personal information on individuals
    - Profiling and nonobvious relationship awareness (NORA)
  - **Networking advances and the Internet**
    - Enables moving and accessing large quantities of personal data

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Ethics in an Information Society

- **Basic concepts form the underpinning of an ethical analysis of information systems and those who manage them**
  - **Responsibility:** Accepting the potential costs, duties, and obligations for decisions
  - **Accountability:** Mechanisms for identifying responsible parties
  - **Liability:** Permits individuals (and firms) to recover damages done to them through law. Liability extends the concept of Responsibility further to the area of law
  - **Due process:** Laws are well known and understood, with an ability to appeal to higher authorities

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Ethics in an Information Society

- **Ethical analysis: A five-step process**
  1. Identify and clearly describe the facts
  2. Define the conflict or dilemma and identify the higher-order values involved
  3. Identify the stakeholders
  4. Identify the options that you can reasonably take
  5. Identify the potential consequences of your options

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- **Candidate Ethical Principles**

1. **Golden Rule:** Do unto others as you would have them do unto you.
2. **Immanuel Kant's Categorical Imperative:** If an action is not right for everyone to take, it is not right for anyone.
3. **Descartes' rule of change:** If an action cannot be taken repeatedly, it is not right to take at all.
4. **Utilitarian Principle:** Take the action that achieves the higher or greater value.
5. **Risk Aversion Principle:** Take the action that produces the least harm or the least potential cost.
6. **no free lunch" rule:** Assume that virtually all tangible and intangible objects are owned by someone else unless there is a specific declaration otherwise.

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Ethics in an Information Society

- **Professional codes of conduct**
  - Promulgated by associations of professionals
    - E.g. AMA, ABA, AITP, ACM
  - Promises by professions to regulate themselves in the general interest of society
- **Real-world ethical dilemmas**
  - One set of interests pitted against another
  - E.g., Right of company to maximize productivity of workers vs. workers right to use Internet for short personal tasks

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- **Information rights and obligations**
  - **Privacy**
    - Claim of individuals to be left alone, free from surveillance or interference from other individuals, organizations, or the state.
    - Ability to control information about yourself
  - In U.S., privacy protected by:
    - First Amendment (freedom of speech)
    - Fourth Amendment (unreasonable search and seizure)
    - Additional federal statues
      - Privacy Act of 1974

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- **Fair information practices:**
  - Set of principles governing the collection and use of information
  - Basis of most U.S. and European privacy laws
  - Based on mutuality of interest between record holder and individual
  - Restated and extended by FTC in 1998 to provide guidelines for protecting online privacy
  - Used to drive changes in privacy legislation
    - COPPA
    - Gramm-Leach-Bliley Act
    - HIPAA

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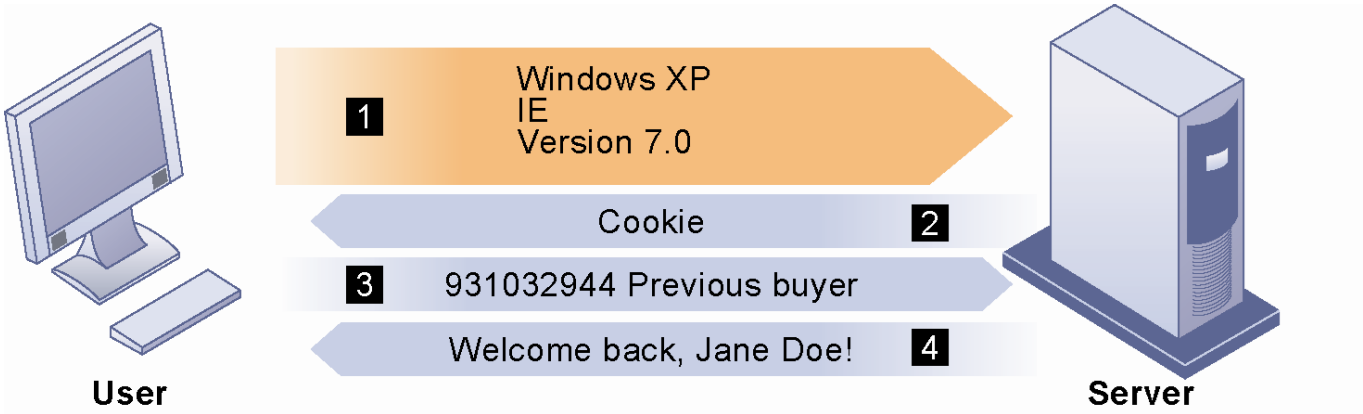
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- **Internet Challenges to Privacy:**
  - **Cookies**
    - Tiny files downloaded by Web site to visitor's hard drive
    - Identify visitor's browser and track visits to site
    - Allow Web sites to develop profiles on visitors
  - **Web bugs**
    - Tiny graphics embedded in e-mail messages and Web pages
    - Designed to monitor who is reading a message and transmitting that information to another computer on the Internet
  - **Spyware**
    - Surreptitiously installed on user's computer
    - May transmit user's keystrokes or display unwanted ads

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How Cookies Identify Web Visitors



1. The Web server reads the user's Web browser and determines the operating system, browser name, version number, Internet address, and other information.
2. The server transmits a tiny text file with user identification information called a cookie, which the user's browser receives and stores on the user's computer hard drive.
3. When the user returns to the Web site, the server requests the contents of any cookie it deposited previously in the user's computer.
4. The Web server reads the cookie, identifies the visitor, and calls up data on the user.

Cookies are written by a Web site on a visitor's hard drive. When the visitor returns to that Web site, the Web server requests the ID number from the cookie and uses it to access the data stored by that server on that visitor. The Web site can then use these data to display personalized information.

Figure 4-3

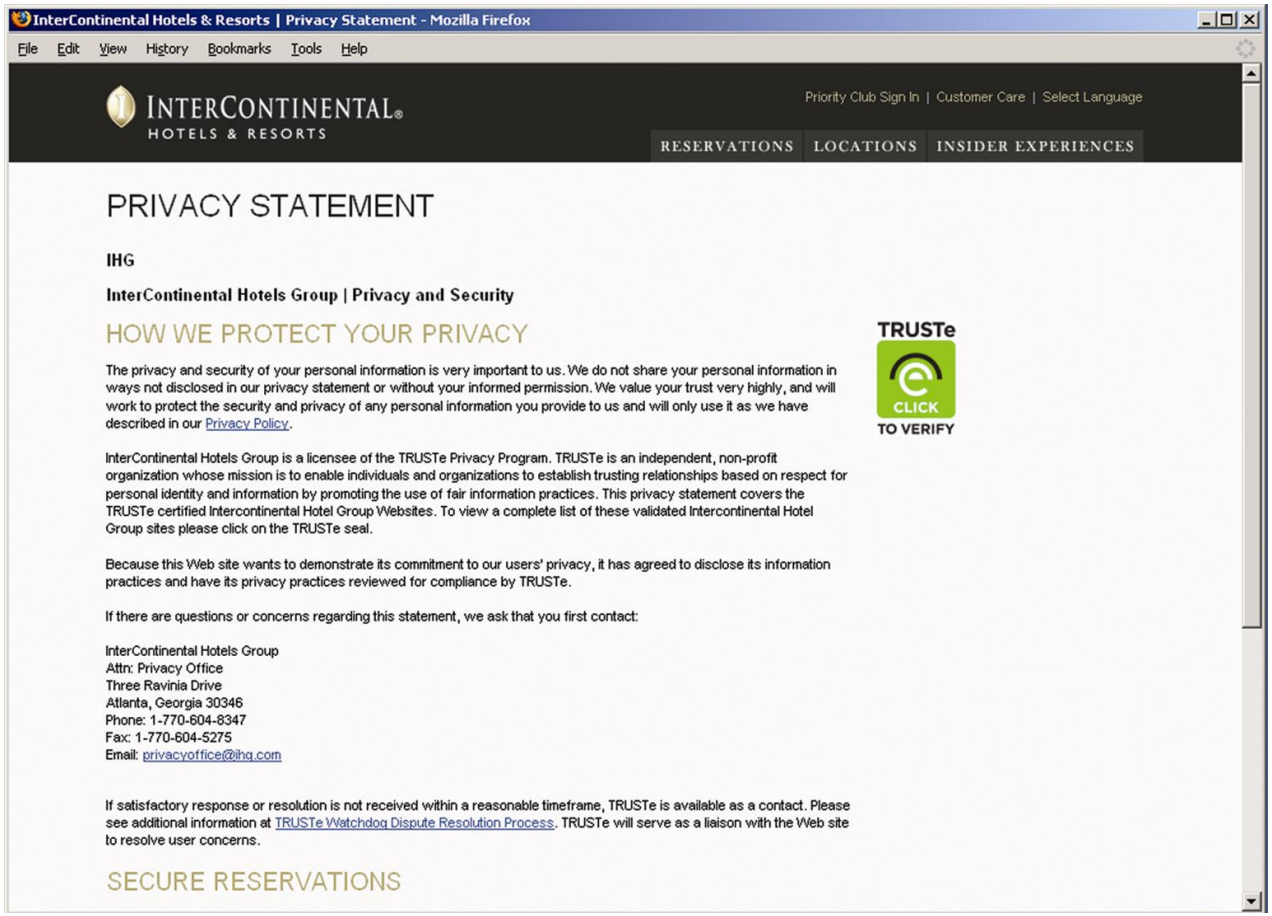
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- U.S. allows businesses to gather transaction information and use this for other marketing purposes
- Online industry promotes self-regulation over privacy legislation
- Self regulation has proven highly variable
  - Statements of information use are quite different
  - Some firms offer opt-out selection boxes
  - Online “seals” of privacy principles
- Most Web sites do not have any privacy policies
- Many online privacy policies do not protect customer privacy, but rather protect the firm from lawsuits

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Web sites are posting their privacy policies for visitors to review. The TRUSTe seal designates Web sites that have agreed to adhere to TRUSTe's established privacy principles of disclosure, choice, access, and security.

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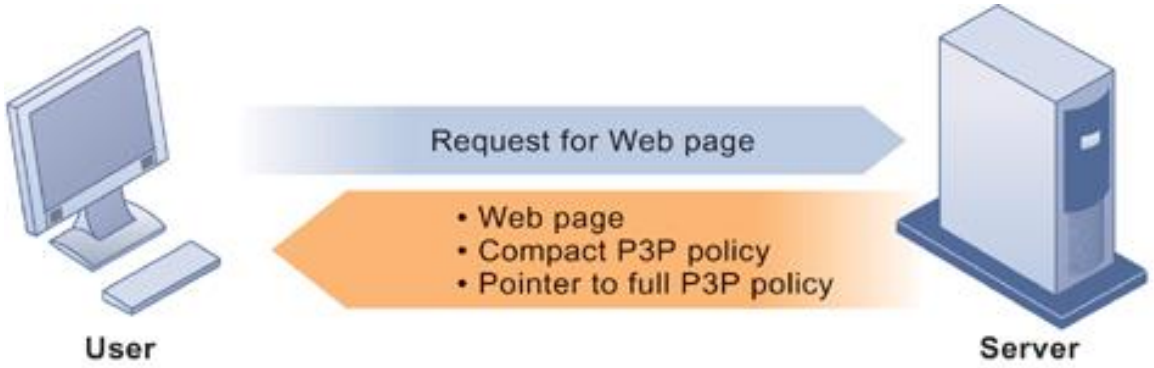
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- **Technical solutions**
  - **The Platform for Privacy Preferences (P3P)**
    - Allows Web sites to communicate privacy policies to visitor's Web browser – user
    - User specifies privacy levels desired in browser settings
    - E.g., "medium" level accepts cookies from first-party host sites that have opt-in or opt-out policies but rejects third-party cookies that use personally identifiable information without an opt-in policy.

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The P3P Standard



1. The user with P3P Web browsing software requests a Web page.
2. The Web server returns the Web page along with a compact version of the Web site's policy and a pointer to the full P3P policy. If the Web site is not P3P compliant, no P3P data are returned.
3. The user's Web browsing software compares the response from the Web site with the user's privacy preferences. If the Web site does not have a P3P policy or the policy does not match the privacy levels established by the user, it warns the user or rejects the cookies from the Web site. Otherwise, the Web page loads normally.

P3P enables Web sites to translate their privacy policies into a standard format that can be read by the user's Web browser software. The user's Web browser software evaluates the Web site's privacy policy to determine whether it is compatible with the user's privacy preferences.

Figure 4-4

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- **Property Rights: Intellectual Property**
  - **Intellectual property:** Intangible property of any kind created by individuals or corporations
  - Three ways that intellectual property is protected
    - **Trade secret**: Intellectual work or product belonging to business, not in the public domain
    - **Copyright**: Statutory grant protecting intellectual property from being copied for the life of the author, plus 70 years. For corporate works, it lasts for 95 years after creation. Drawback is that only work is protected, not the idea
    - **Patents**: Grants creator of invention an exclusive monopoly on ideas behind invention for 20 years

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- **Challenges to Intellectual Property Rights**
  - Digital media different from physical media (e.g. books)
    - Ease of replication
    - Ease of transmission (networks, Internet)
    - Difficulty in classifying software
    - Compactness
    - Difficulties in establishing uniqueness
- **Digital Millenium Copyright Act (DMCA)**
  - Makes it illegal to circumvent technology-based protections of copyrighted materials

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- **Accountability, Liability, Control**
  - Computer-related liability problems
    - If software fails, who is responsible?
      - If seen as a part of a machine that injures or harms, software producer and operator may be liable
      - If seen as similar to a book, difficult to hold software author/publisher responsible
      - What should liability be if software is seen as service? Would this be similar to telephone systems not being liable for transmitted messages (so-called “common carriers”)

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- **System Quality: Data Quality and System Errors**
  - What is an acceptable, technologically feasible level of system quality?
    - Flawless software is economically unfeasible
  - Three principal sources of poor system performance:
    - Software bugs, errors
    - Hardware or facility failures
    - Poor input data quality (most common source of business system failure)

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- **Quality of Life:** Negative social consequences of systems
  - **Balancing power**: Although computing power is decentralizing, key decision-making power remains centralized
  - **Rapidity of change**: Businesses may not have enough time to respond to global competition
  - **Maintaining boundaries**: Computing and Internet use lengthens the work-day, infringes on family, personal time
  - **Dependence and vulnerability**: Public and private organizations ever more dependent on computer systems

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What Should We Do About Cyberbullying?

- **Read the Interactive Session: Organizations, and then discuss the following questions:**
  - **What are some of the technologies and communication methods used by cyberbullies? Why are they so effective?**
  - **What measures have been taken by school districts and governments to combat cyberbullying? How helpful are they? Can you think of any other ways to effectively control cyberbullying?**
  - **Should there be stronger laws outlawing cyberbullying? Why or why not?**
  - **Does a social networking site catering to teems such as MySpace represent an ethical dilemma? Why or why not?**

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- **Computer crime and abuse**
  - **Computer crime**: Commission of illegal acts through use of compute or against a computer system – computer may be object or instrument of crime
  - **Computer abuse**: Unethical acts, not illegal
    - **Spam**: High costs for businesses in dealing with spam
- **Employment:** Reengineering work resulting in lost jobs
- **Equity and access – the digital divide**: Certain ethnic and income groups in the United States less likely to have computers or Internet access

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Flexible Scheduling at Wal-Mart: Good or Bad for Employees?

- **Read the Interactive Session: Management, and then discuss the following questions:**
  - **What is the ethical dilemma facing Wal-Mart in this case? Do Wal-Mart's associates also face an ethical dilemma? If so, what is it?**
  - **What ethical principles apply to this case? How do they apply?**
  - **What are the potential effects of computerized scheduling on employee morale? What are the consequences of these effects for Wal-Mart?**

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- **Health risks:**
  - Repetitive stress injury (RSI)
    - Largest source is computer keyboards
    - Carpal Tunnel Syndrome (CTS)
  - Computer vision syndrome (CVS)
  - Technostress
  - Role of radiation, screen emissions, low-level electromagnetic fields

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